



Important Formulas & Points to Remember: Another Beyond the Point

Table of Contents

1. What Are Decimals?
2. Dividing by 10, 100, 1000...
3. Rule for Multiplying Two Decimals
4. Is the Product Always Greater?
5. Decimal Division by Place Value

What Are Decimals?

Decimals are a way of writing numbers that include **whole parts and fractional parts** together.

A decimal number has **two parts**:

Whole-number part

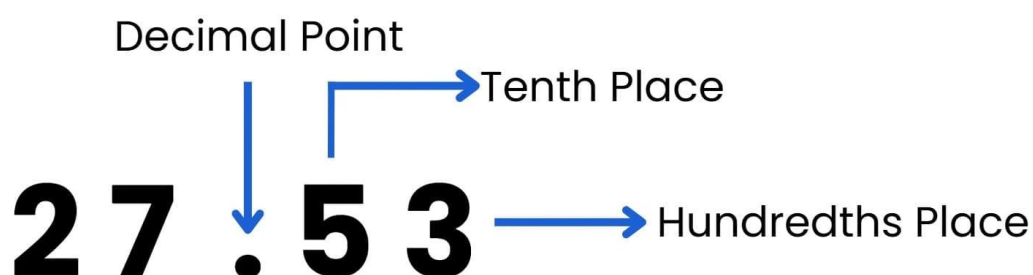
Left of the decimal point

Example: In 27.53 → 27 is whole part

Fractional part

Right of the decimal point

Example: In 27.53 → 53 is fractional part



**Each digit has a place value:**

Tenths $\rightarrow 1/10$

Hundredths $\rightarrow 1/100$

Thousandths $\rightarrow 1/1000$

Example :

27.53 =

2 tens = 20

7 ones = 7

5 tenths = 0.5

3 hundredths = 0.03

So,

$$27.53 = 20 + 7 + 0.5 + 0.03$$

Understanding this breakdown helps in multiplication and division.

Dividing by 10, 100, 1000...

To divide by 10, 100, 1000...

Move the decimal point to the LEFT

...by as many places as the number of zeros.

Examples:

$$123 \div 10 = 12.3$$

$$123 \div 100 = 1.23$$

$$123 \div 1000 = 0.123$$

If the number doesn't have enough digits, put **0s** in front.

This is the **foundation** for decimal division.

Rule for Multiplying Two Decimals

Steps:

1. Remove decimal points
2. Multiply as whole numbers
3. Count total decimal places
4. Place decimal from right accordingly

Example: 5.96×24.8

Decimal places = $2 + 1 = 3$

Whole number multiplication: $596 \times 248 = 147808$

Final answer: **147.808**

$$\begin{array}{rcccl}
 & & 596 \times 248 & = & 147808 \\
 & \underbrace{}_{2 \text{ decimal places}} \times \underbrace{}_{1 \text{ decimal place}} & & & \underbrace{}_{3 \text{ decimal places}} \\
 & \underbrace{}_{2 + 1} & & & \\
 & = 3 \text{ decimal places} & & &
 \end{array}$$

Is the Product Always Greater?

No

Case 1: Product > both numbers

$$2.25 \times 8 = 18$$

Case 2: Product between the two numbers

$$0.25 \times 8 = 2$$

Case 3: Product < both numbers

$$0.25 \times 0.8 = 0.20$$

Note:

If both decimals are **less than 1**, the product becomes **smaller**.

If one is greater and one is less than 1, product lies **between** them.

If both are large, product becomes **large**.

Decimal Division by Place Value

When dividing whole numbers and decimals:

Use long division

Diagram illustrating the long division of 237.025 by 8, showing the quotient and the corresponding place values:

	H	T	O	Tenths	Hundredths	Thousandths
8) 237 (0 2 9 . 6 2 5						
16						
77						
72						
50						
48						
20						
16						
40						
40						
0						

Place value labels and arrows:

- Tens
- Ones
- Tenths
- Hundredths
- Thousandths

Place decimal in quotient exactly above its position in dividend

01

Writing the division as a fraction.

Changing the fraction into an equivalent form whose denominator is 1, 10, 100, 1000, ...

02

03

Writing the result as a decimal.

